Expt. No. 5

Title: To design layout of NAND and NOR gate using micro-wind.

Rubric	*Score (1 to 5)*
Timely Submission	
Result Interpretation and Output	
Program Logic and Concept Understanding	
Total	

Performed On: _____

Sign:_____

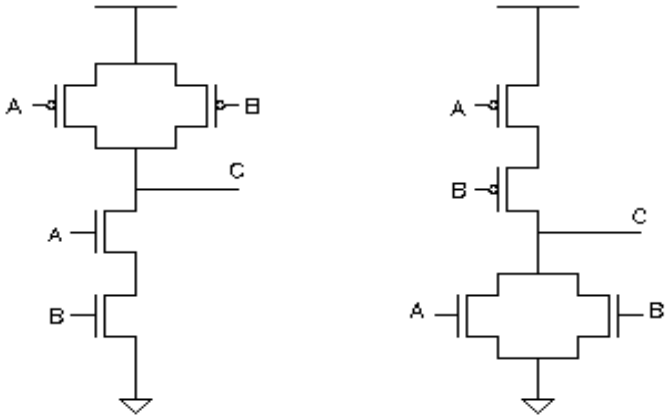
Experiment No. 5

Aim: To design layout of NAND and NOR gate using micro-wind.

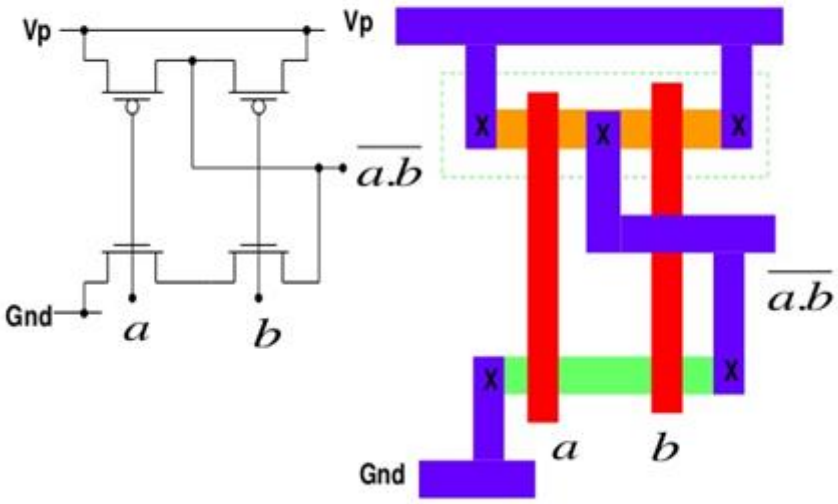
Software Required: Micro-wind software, DSCH software

Hardware: computer

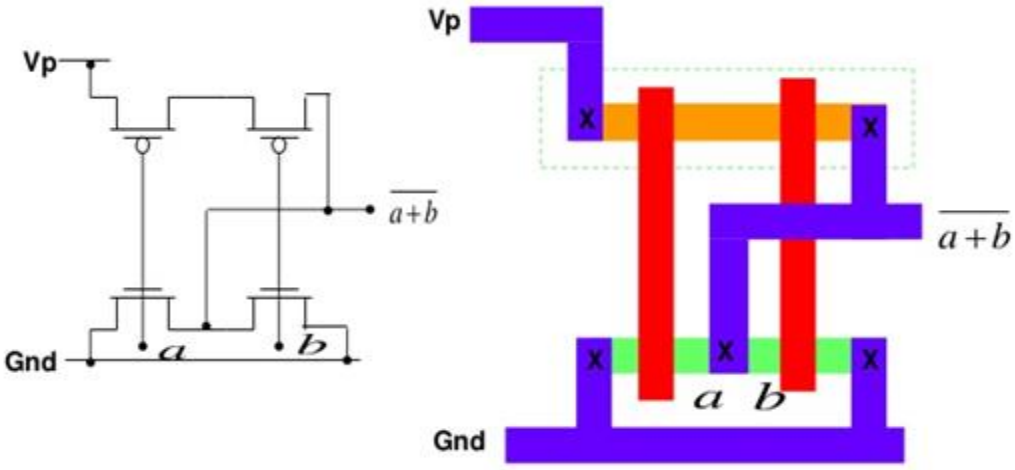
Circuit Diagram:



The CMOS NAND Gate



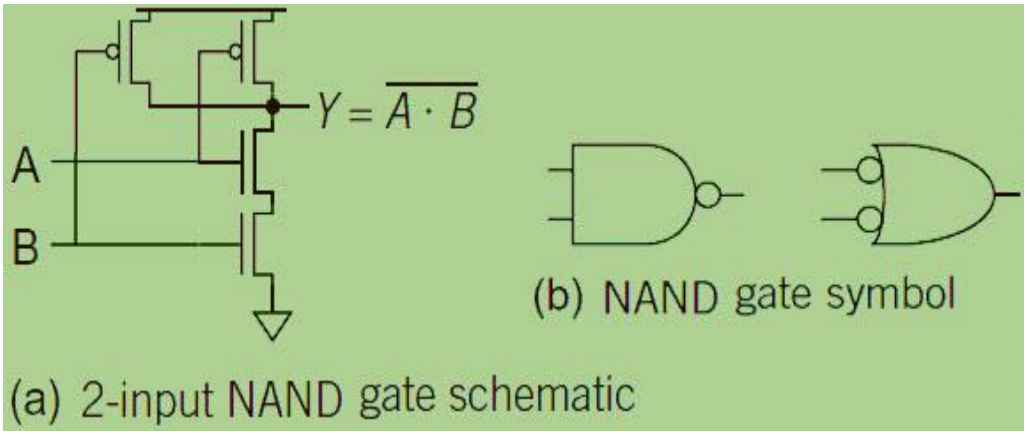
The CMOS NOR Gate



Theory:

CMOS NAND Gate

The below figure shows a 2-input Complementary MOS NAND gate. It consists of two series NMOS transistors between Y and Ground and two parallel PMOS transistors between Y and VDD.



CMOS NAND Gate

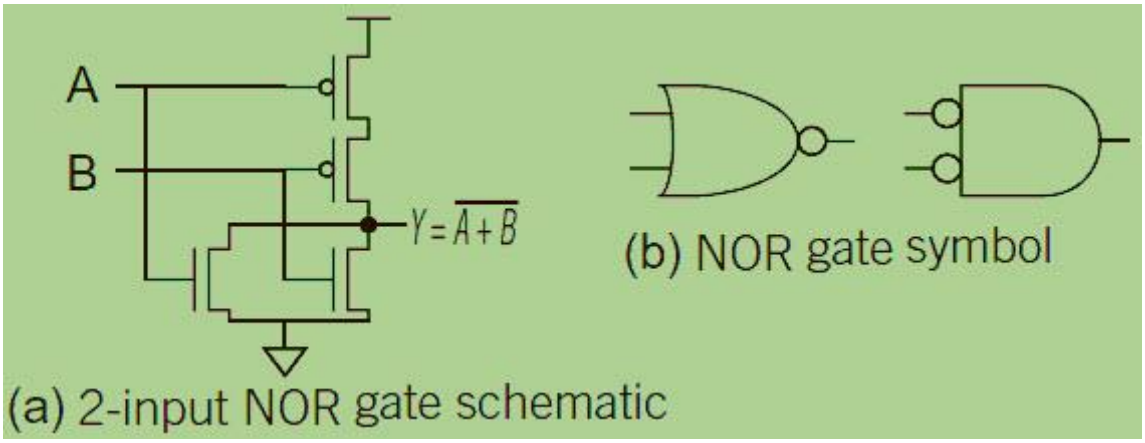
If either input A or B is logic 0, at least one of the NMOS transistors will be OFF, breaking the path from Y to Ground. But at least one of the pMOS transistors will be ON, creating a path from Y to VDD.

Hence, the output Y will be high. If both inputs are high, both of the NMOS transistors will be ON and both of the PMOS transistors will be OFF. Hence, the output will be logic low. The truth table of NAND logic gate given in below table.

A	B	Pull-Down Network	Pull-up Network	OUTPUT Y
0	0	OFF	ON	1
0	1	OFF	ON	1
1	0	OFF	ON	1
1	1	ON	OFF	0

CMOS NOR Gate

A 2-input NOR gate is shown in the figure below. The NMOS transistors are in parallel to pull the output low when either input is high. The PMOS transistors are in series to pull the output high when both inputs are low, as given in below table. The output is never left floating.



The truth table of NOR logic gate given in below table.

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

NAND and NOR gate can be implemented using CMOS inverter. It consists of NMOS and PMOS two transistors and they have to be proved in a layout such that we get device output according to the logic. Some important conditions to be kept in mind during layout design are

1. N well/ P well spacing =6λ
2. N well/ P well width =6λ
3. Active spacing = 3λ
4. Poly width = 3λ
5. Metal width = 2λ
6. Metal to metal = 2λ
7. Mask = 3λ

Procedure: DSCH software

1. Open DSCH software.
2. Do the connections of circuit in software as per circuit diagram.
3. Check the connections before running it. If not connected, do the connections without any open circuit.
4. Save & Run the software.
5. See the simulated waveforms.
6. Check whether the waveforms follow the desired results
7. Record the waveform obtained.

Procedure: Micro-wind software

1] Start micro-wind software for layout**.**

2] Draw the CMOS inverter following the design.

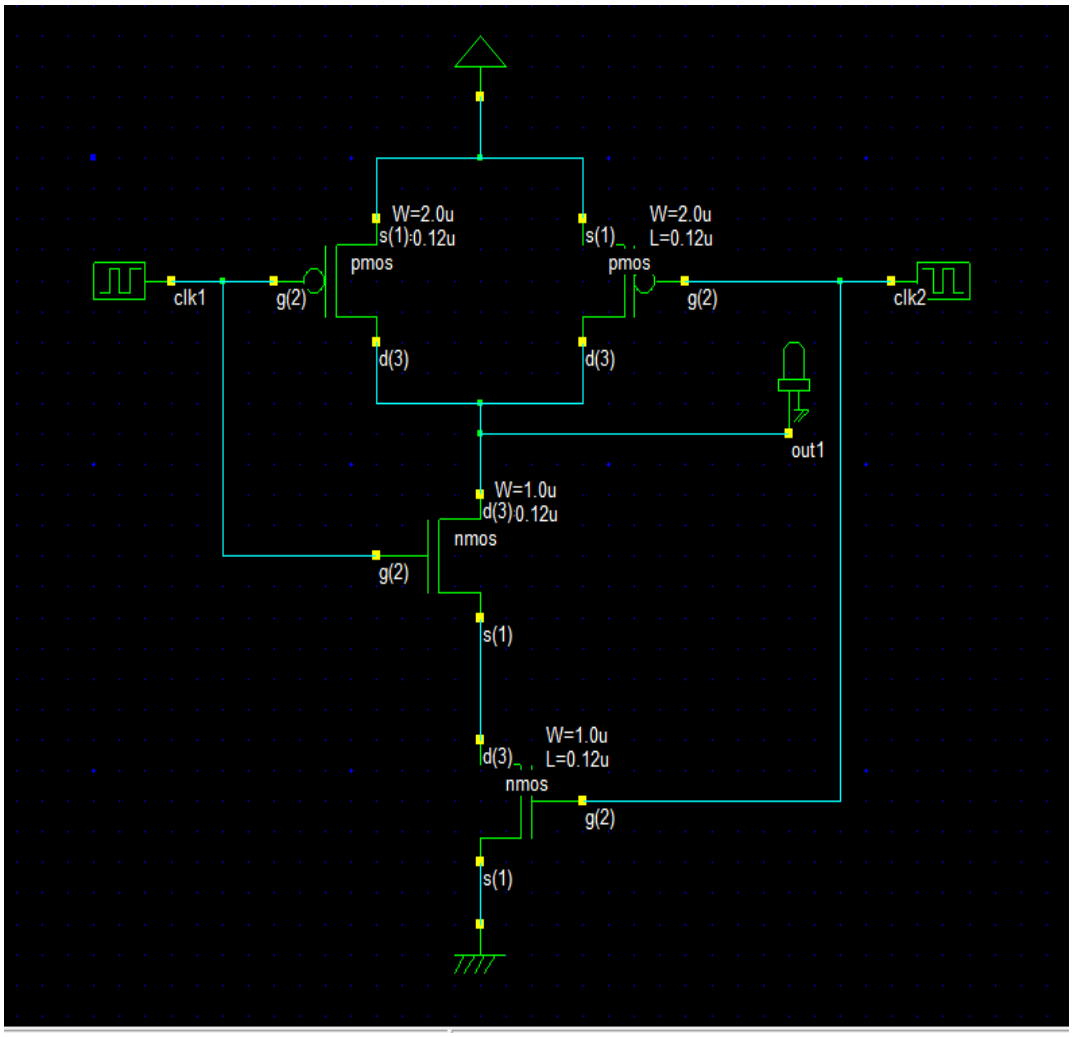
3] Click on DRC at every stage of design to avoid error.

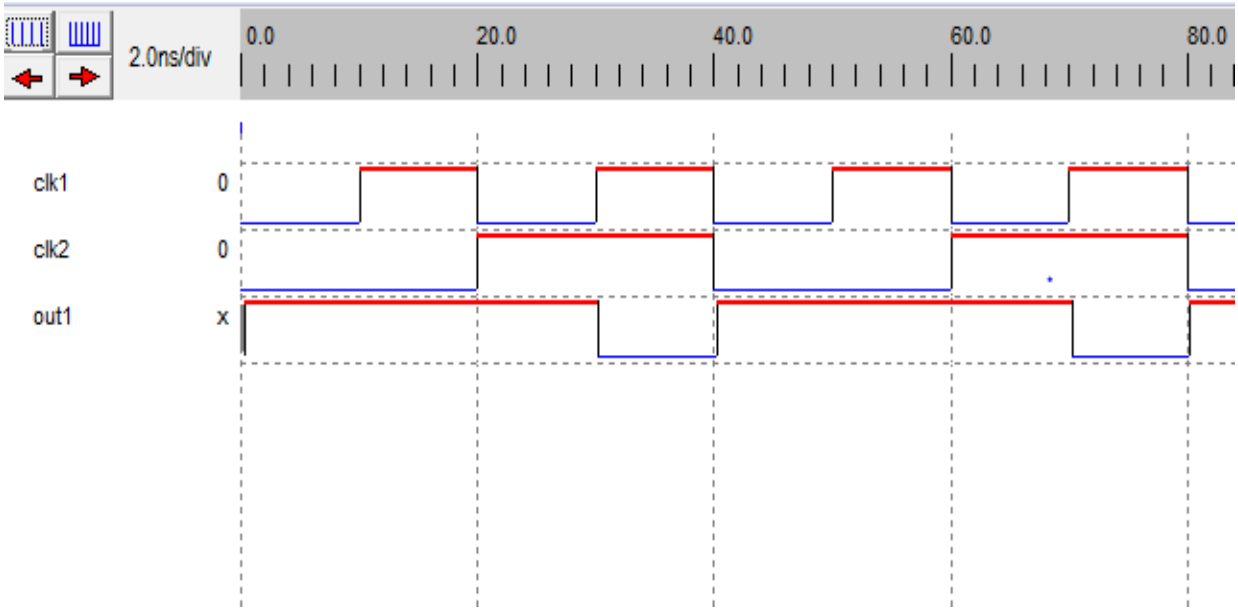
4] Observe various output while using simulation output.

5] Software design and finally run a DRC.

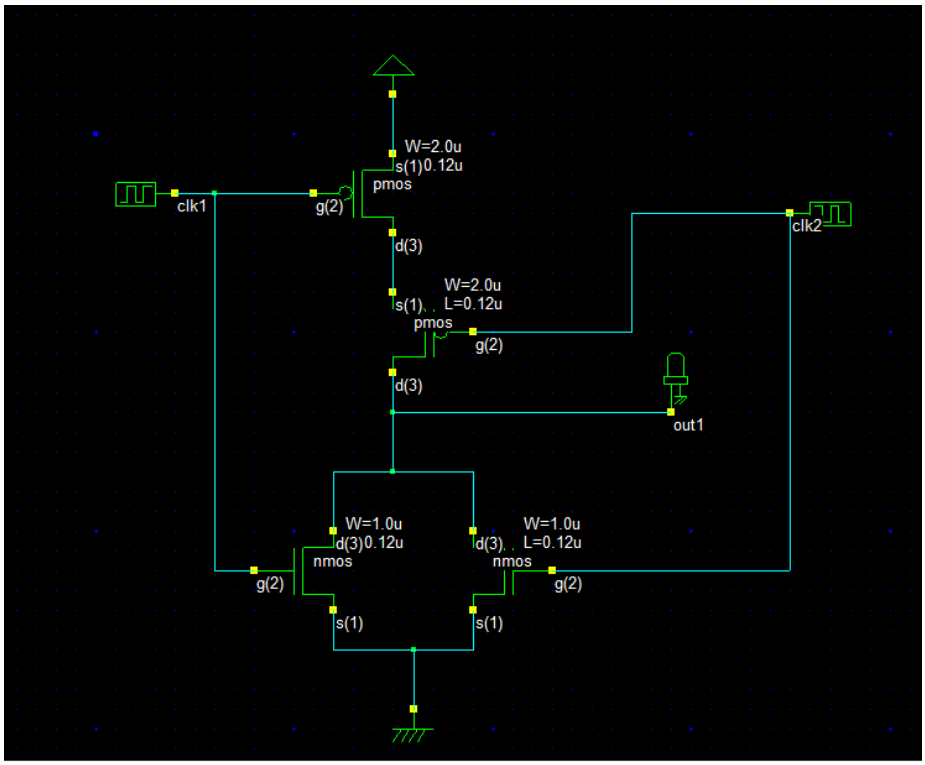
6] Record the output waveform observed

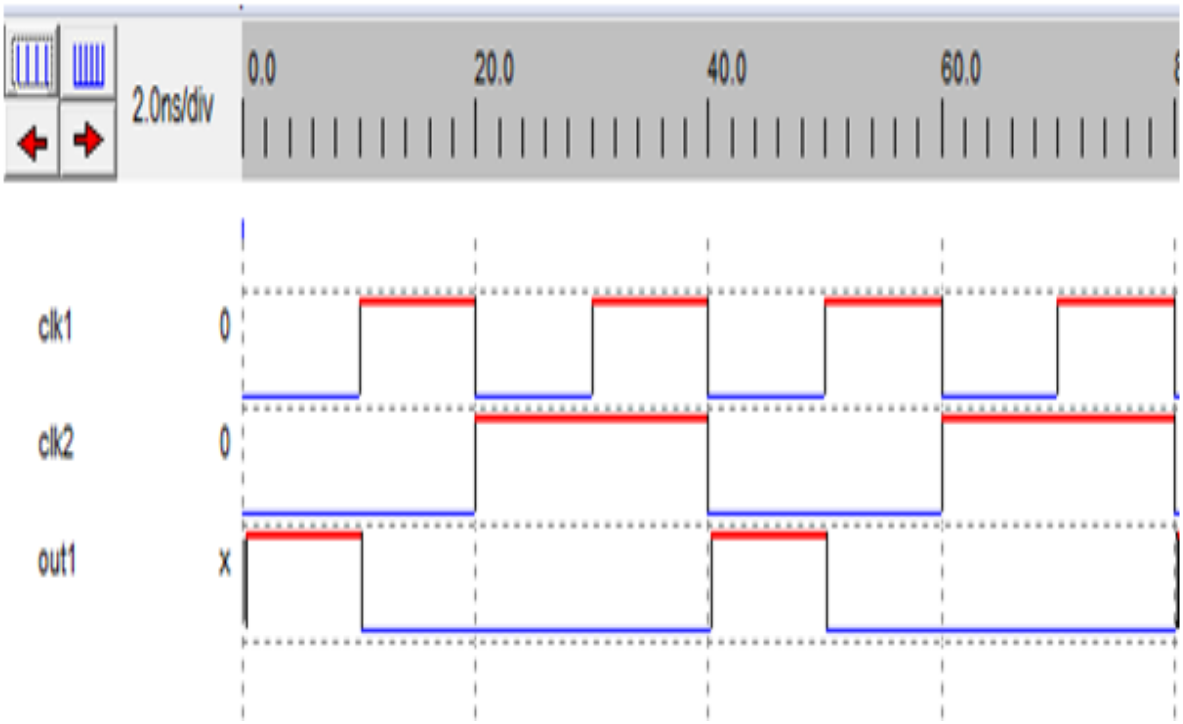
Output (NAND circuit):



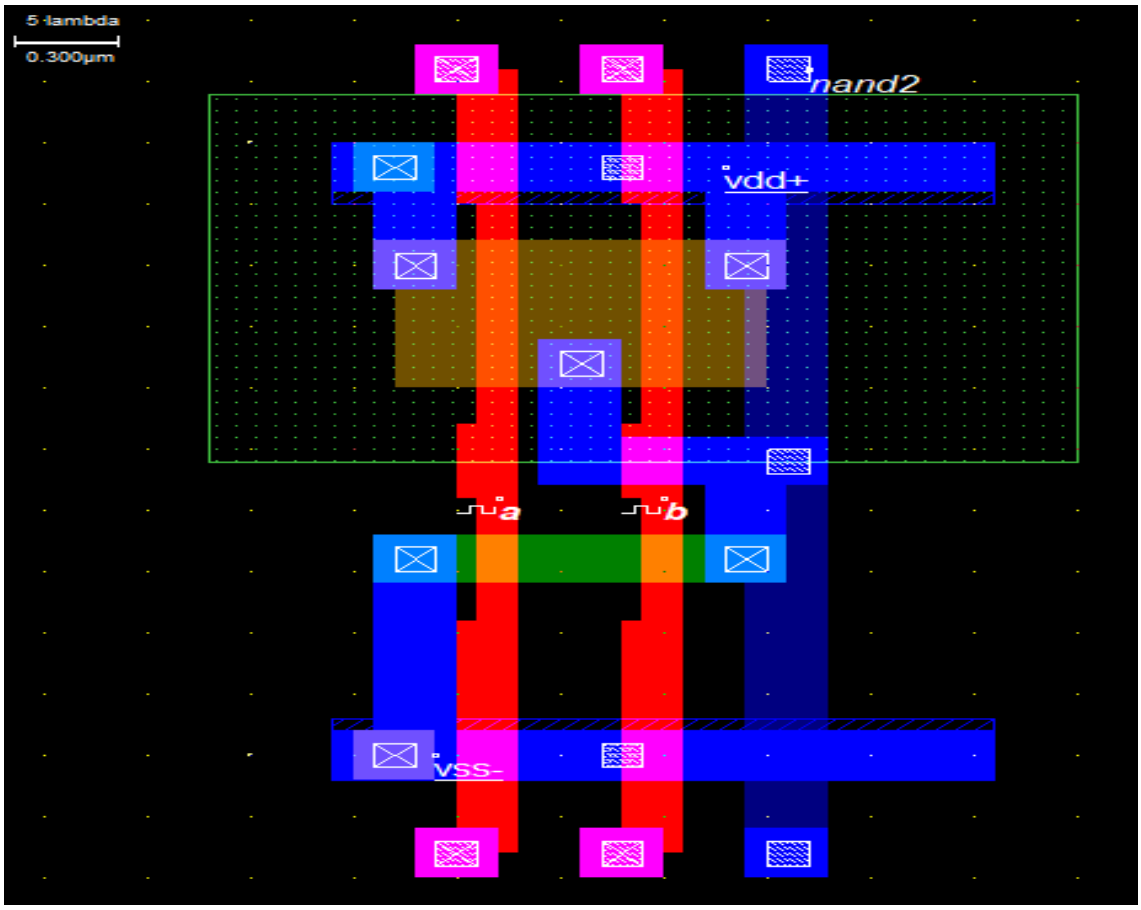


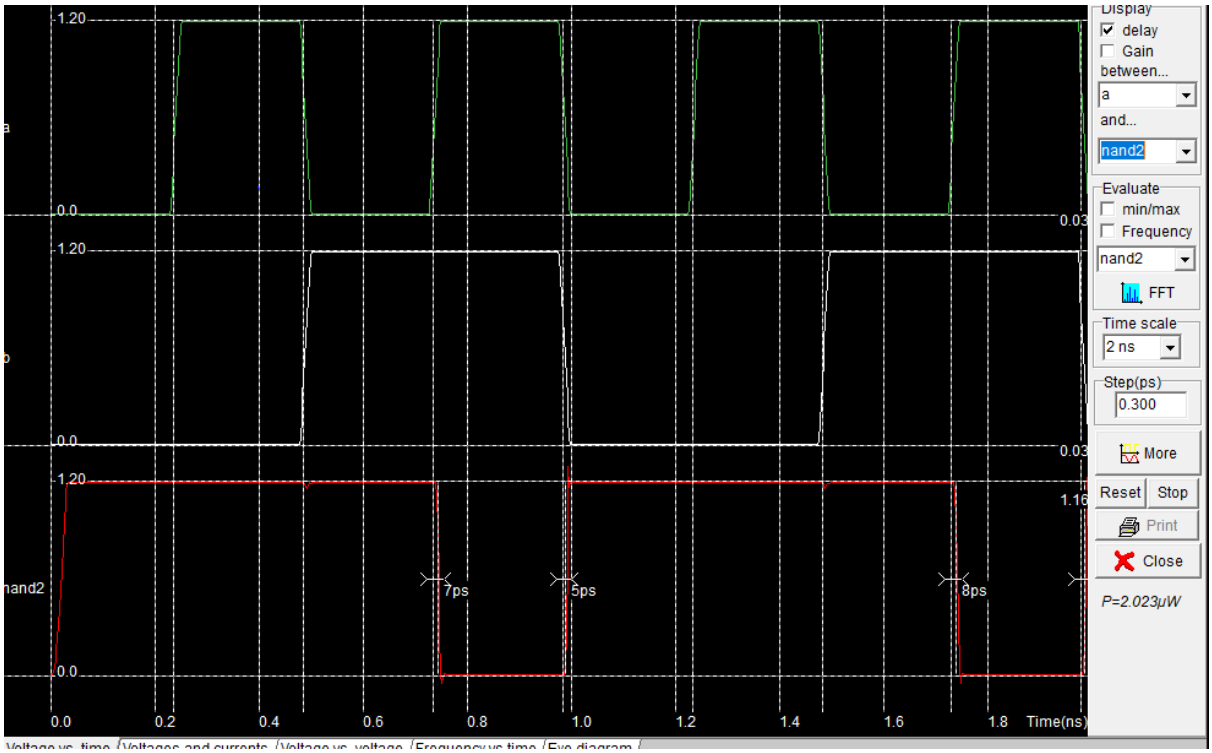
Output (NOR circuit):

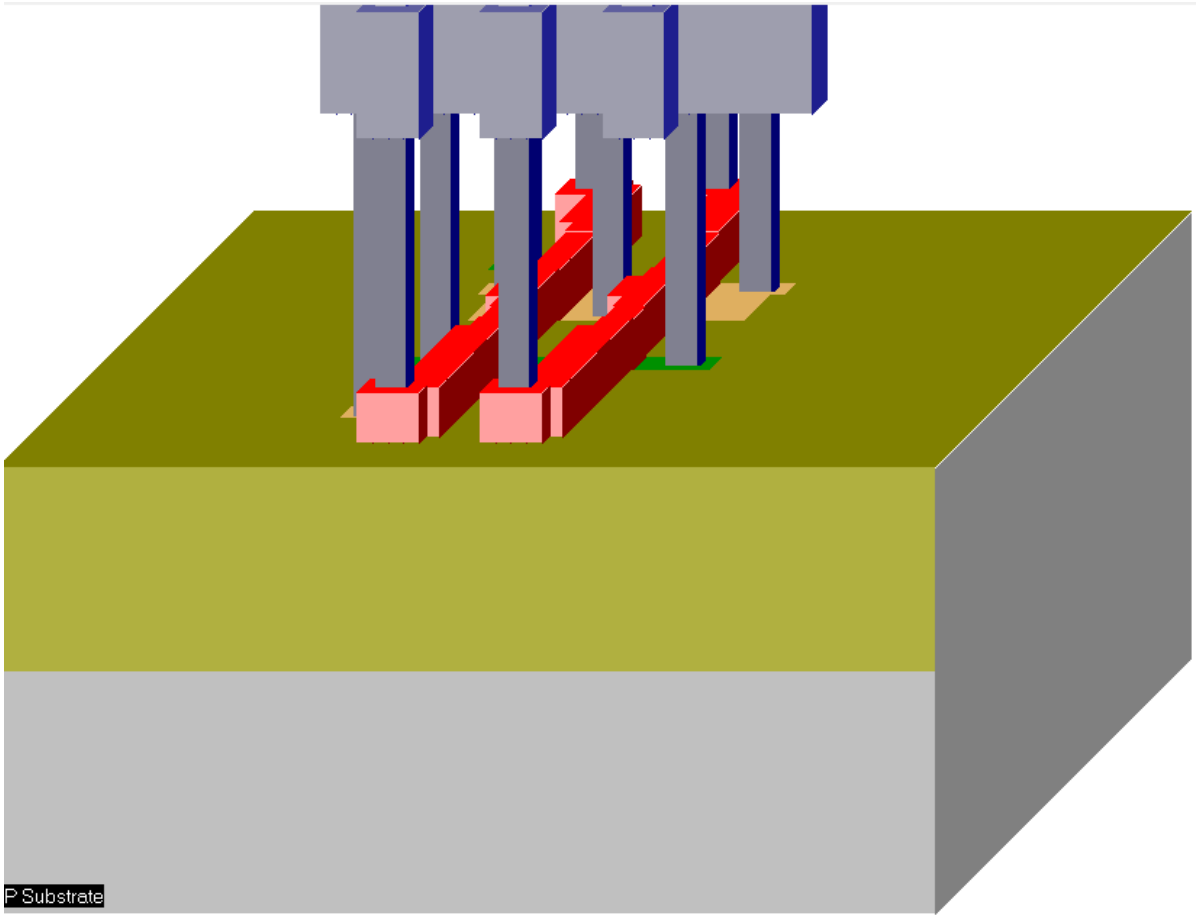




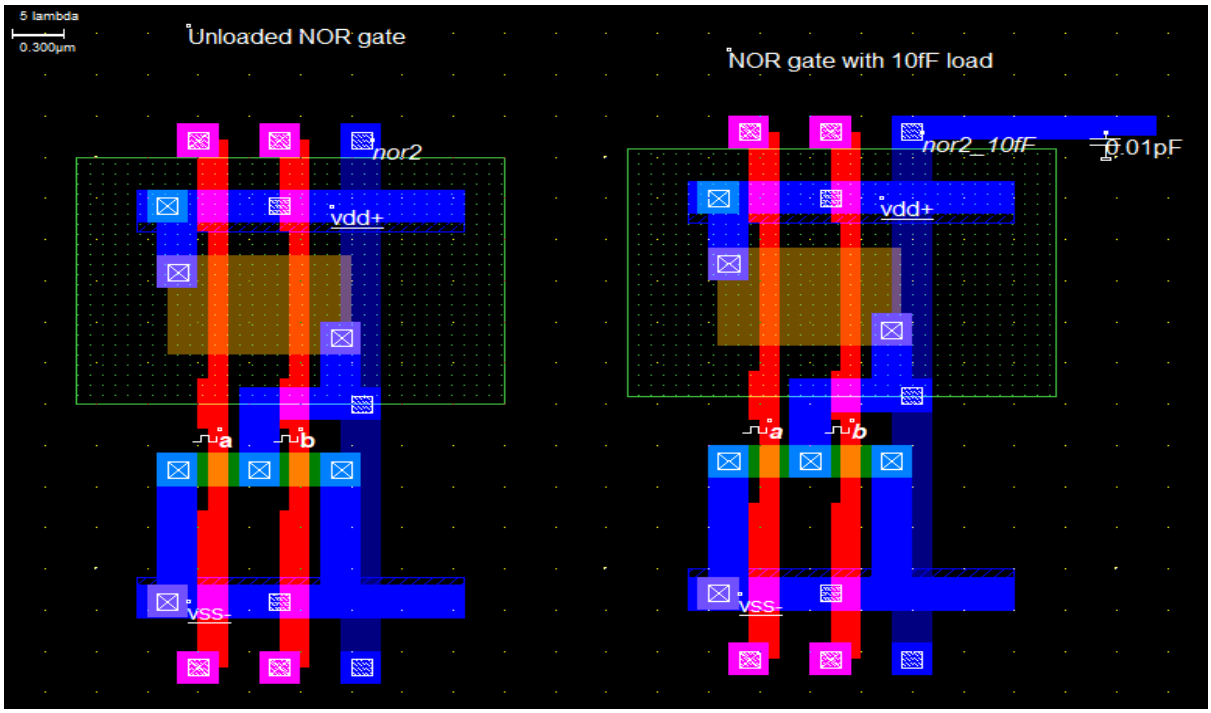
Layout NAND circuit:

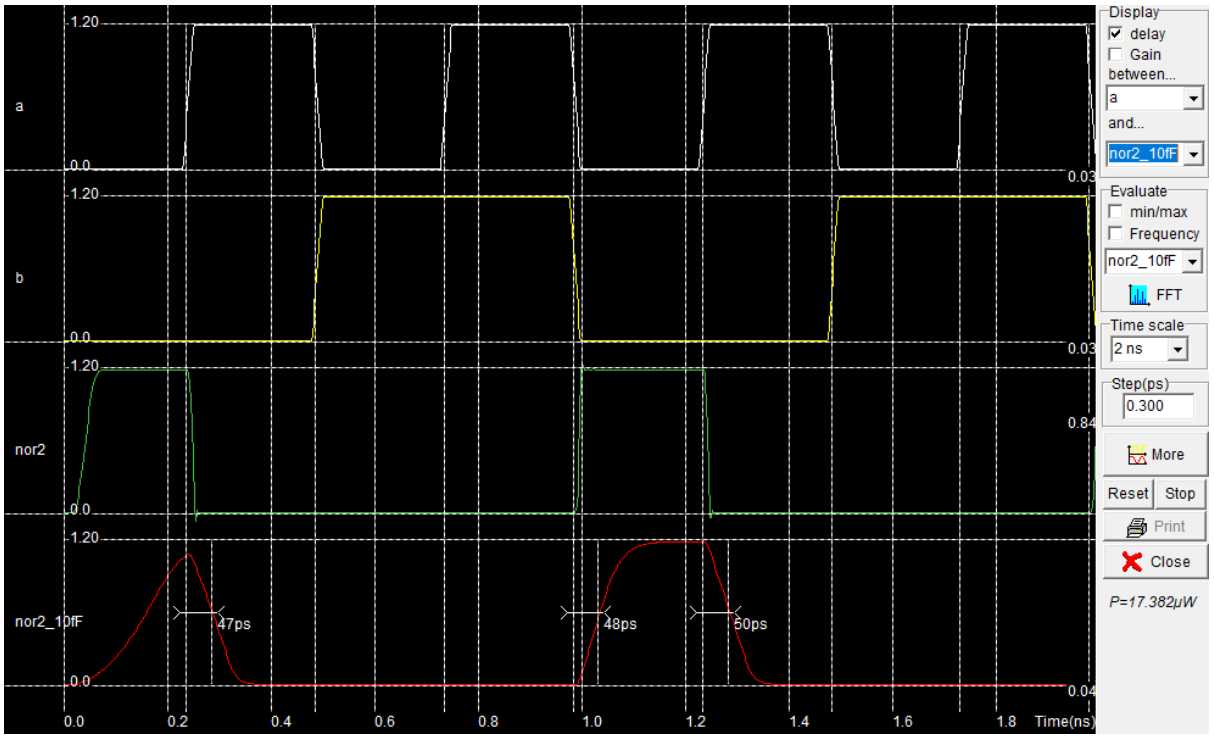


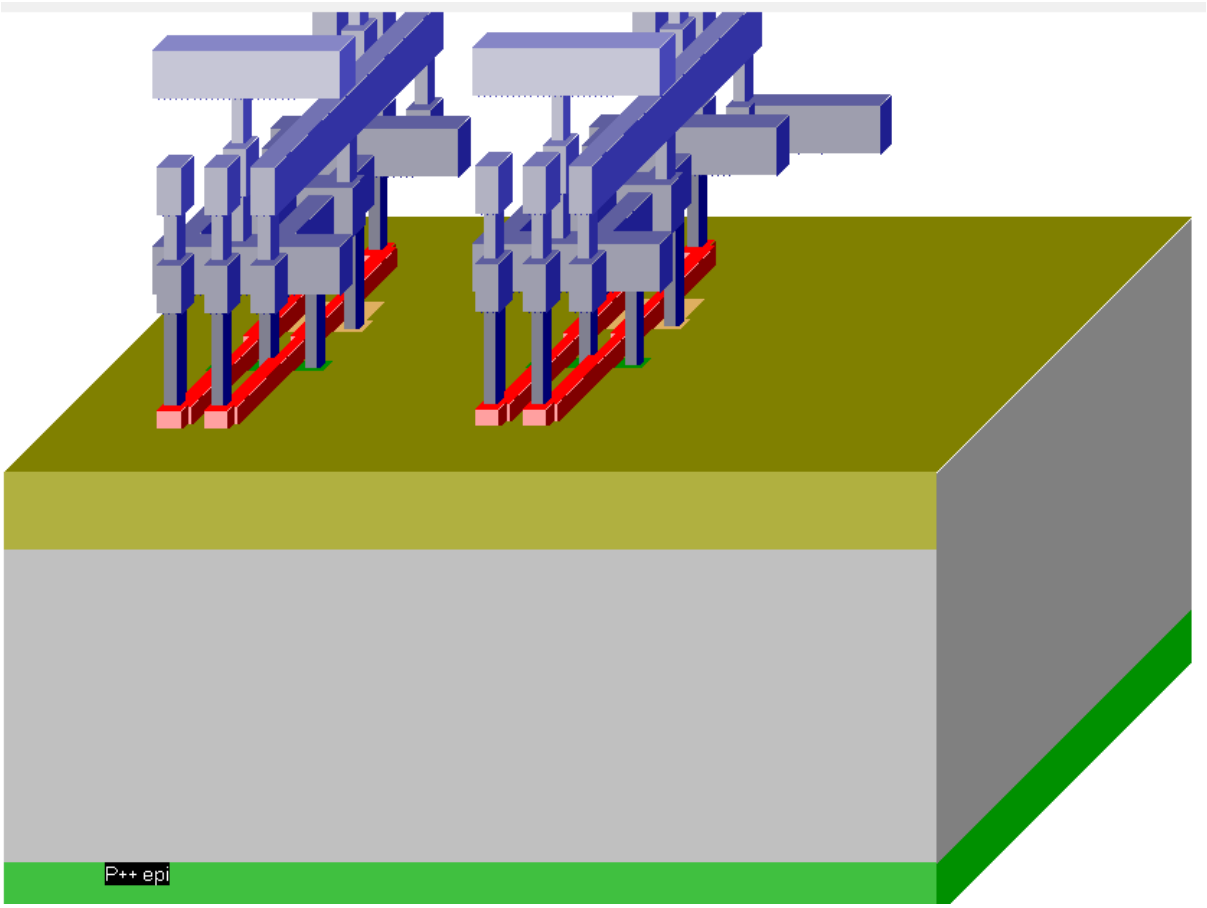




Layout NOR circuit:




a
b
nor2
nor2_10fF
1.20
0.0
47ps
48ps
50ps
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
Time(ns)
Display
delay
Gain
between...
and...
Evaluate
min/max
Frequency
FFT
Time scale
2 ns
Step(ps)
0.300
More
Reset
Stop
Print
Close
P=17.382μW


P++ epi

Pre Lab Questions:

1 Why inputs of a NAND gate are connected together?
2 Why inputs of a NOR gate are connected together?

Post Lab Questions:

1. How the NAND occupy less area than the NOR?

Conclusion: Thus, we have designed the layout of NAND and NOR gates as per the design rules of micro wind and observed the output.

CO's Covered:

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